DC-FOUNDATION-01

Raspberry Pi Node Foundation

Lab Execution Record & Meta Report

Project ID	P00 (Foundation) — first executable of **P01, Jan–Apr 2026**
Node	`dc-node-01` (D-Central node 01)
Platform	Raspberry Pi 5 Model B Rev 1.1 (8 GB)
Operating system	Raspberry Pi OS 64-bit — Debian 13 "Trixie"
Kernel	6.18.34+rpt-rpi-2712 (`arm64`)
Author	Redji Jean Baptiste (Toussaint) — 041-269-022
Standard	Master Timeline V22 — Black Start Augmentation System
STDS layer	0 — Userland (activates P01)
Build window	2026-07-09 → 2026-07-22

STATUS: COMPLETE. Part A steps 0–12 executed, evidenced, and verified. SSH hardening proven by negative test. Reboot persistence confirmed. All nine defects (D1–D9) closed. Published to `RedjiJB/cst-portfolio`.

Three-Track status: CL ✓ **PASS** → SL ✓ **PASS** → BSL ✓ **PASS** (rebuild demonstrated). Maturity attained: **BSL-1**, touching **BSL-2**.

Companion research seed **T1.1** — *Minimum Viable Sovereign Infrastructure Specification* → repo `Sovereign-Bootstrap-v1`.

Contents

1 Timeline Alignment — Conformance to the BSA Standard

This report is the first executed artifact of the Master Timeline V22. It is therefore scored not only for technical correctness but against the **Black Start Augmentation System (BSA)** documentation standard: the Three-Track Rule, the six Operational Domains, BSL maturity, the four Operational Modes, and the seven-stage Skill Pipeline. This section maps the P00 work onto that framework so the remainder of the document can be read as evidence for each claim.

1.1 The Three-Track Rule

No skill counts as complete until it clears all three tracks.

Track	Requirement / evidence in this build	Status
CL — Cognitive	Theory of hardening, addressing, and container isolation is explained throughout Parts A/B — e.g. the `sshd_config` include-ordering rule (§16.2) and DHCP-vs-static lease semantics (§2).	✓ **PASS**
SL — System	A working node exists: hardened SSH, host firewall, intrusion protection, automated patching, and a container substrate, all verified live.	✓ **PASS**
BSL — Black Start	Rebuild-from-zero was not hypothetical: a self-inflicted network lockout (§5.2) forced a full re-image, and the node was reconstituted from the documented procedure. The changed Machine ID is the proof (§5.3).	✓ **PASS**

1.2 Operational Domains — P01 target: score 1 in each

Domain	P00 contribution	Score
ENERGY	Out of scope for P00; UPS/solar deferred to later P01 tasks.	0→pending
COMPUTE	Docker substrate operational; `arm64v8` image path proven (Fig. 43).	1
STORAGE	Portfolio vault on Git; artifacts committed; microSD boot (endurance noted §2.2).	1
NETWORK	Static addressing achieved; DNS corrected; firewall default-deny.	1
CONTROL	Key-only SSH proven by negative test; least-privilege user model.	1
PRODUCTION	Reproducible build proven by rebuild; Ansible-style repeatability is the next step.	1

Five of six domains reach the P01 baseline of 1. ENERGY is the single deferred item and is scheduled with the Kill-a-watt power baseline and 2-node cluster later in P01.

1.3 BSL Maturity attained

Phase target for P01–P04 is **BSL-1**/**BSL-2**. This build attains **BSL-1** (lab-capable) across all in-scope domains and demonstrates elements of **BSL-2** (offline/recoverable): the node was recovered from bare storage using documentation alone.

1.4 Four Operational Modes

Mode	Demonstrated?	Status
1 — Online	Normal operation over Wi-Fi / Ethernet.	✓ PASS
2 — Degraded	Node traversed three networks (§5.5); operated on a phone hotspot.	partial
3 — Offline	Not yet — updates still require internet; local mirror is a later task.	pending
4 — Black Start	The lockout → re-image → reconstitution is a partial Mode-4 event (no net, storage rebuilt).	partial

1.5 Seven-Stage Skill Pipeline

LEARN	BUILD	INSPECT	BREAK	DEFEND	RECOVER	OPTIMIZE
✓ PASS	✓ PASS	✓ PASS	✓ PASS	✓ PASS	✓ PASS	partial

Inspect = baseline capture and the `sshd -T` audit; Break = the wrong-subnet lockout and the negative auth test; Recover = the full rebuild. All seven stages are represented, with Optimize carried forward (SSD migration, DHCP reservation).

1.6 Black Start Score Matrix — current node state

Domain	Current state	Target (100)
Energy	Grid only (0)	72 hr battery+solar
Compute	Docker local; rebuild ∼1 image pass	<2 hr full rebuild
Storage	Git vault + microSD (no M-DISC yet)	3-2-1 local + M-DISC
Network	Static + firewall; single path	3 independent paths
Control	Key-only SSH, local user; no cloud IdP	zero cloud dependency
Production	Reproducible (∼1 image pass)	<4 hr from blank HW

Rule satisfied: hard external dependencies were reduced — password authentication and any cloud-IdP path were removed; the node authenticates locally by key with no external identity provider.

1.7 Research seed

This document is the deliverable for research seed **T1.1 — Minimum Viable Sovereign Infrastructure Specification**, and the `cst-portfolio` repository is the concrete instance of `Sovereign-Bootstrap-v1`. The node hostname `dc-node-01` designates it as the first node of **D-Central**.

Part 0

Image Preparation and First Contact

1.8 Tooling

Tool	Version	Role
Raspberry Pi Imager	v2.0.10 (Win 11)	Writes image + first-boot customisation
PuTTY	current	Initial SSH client
OpenSSH (Windows)	built-in	`ssh`/`scp`/`ssh-keygen` from PowerShell

1.9 Device, OS, and storage selection

Raspberry Pi 5 was selected (BCM2712 boot chain). The recommended **Raspberry Pi OS 64-bit (Trixie)** image was used — a deliberate deviation from the manual's Bookworm target, since Bookworm is now the *Legacy* entry and Trixie receives current security updates (analysed in §16.1).

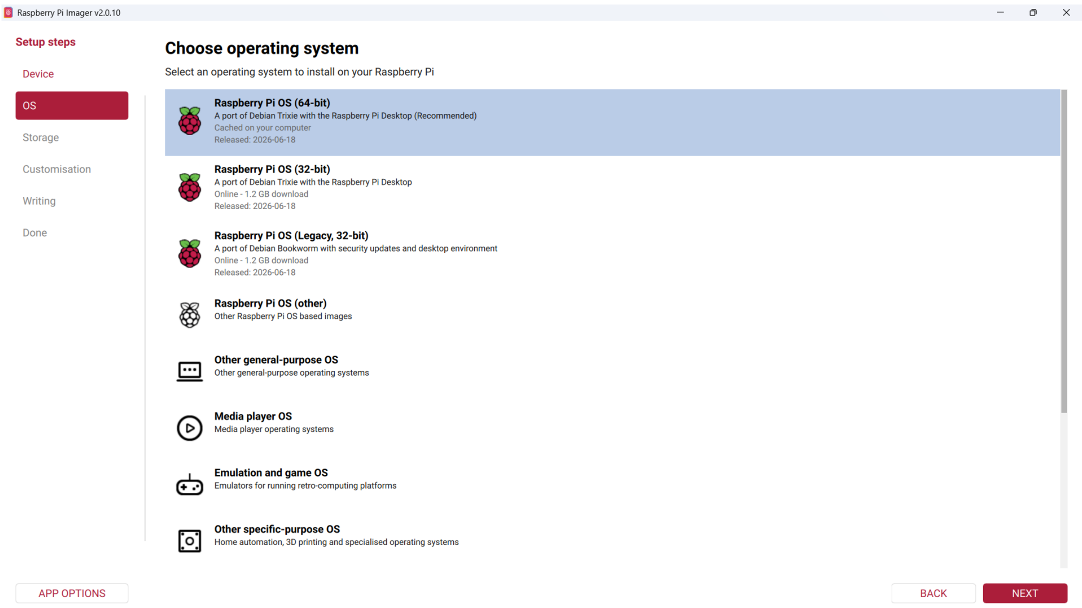


Figure 1: Imager — Raspberry Pi OS 64-bit (Trixie), released 2026-06-18, cached locally.

The only eligible target was a 59.5 GB "USB Mass Storage Device." A Windows AutoPlay prompt claimed the drive was damaged; this is a false positive — Windows cannot read the Linux `ext4` partition and must be ignored (never run the repair tool on a fresh image).

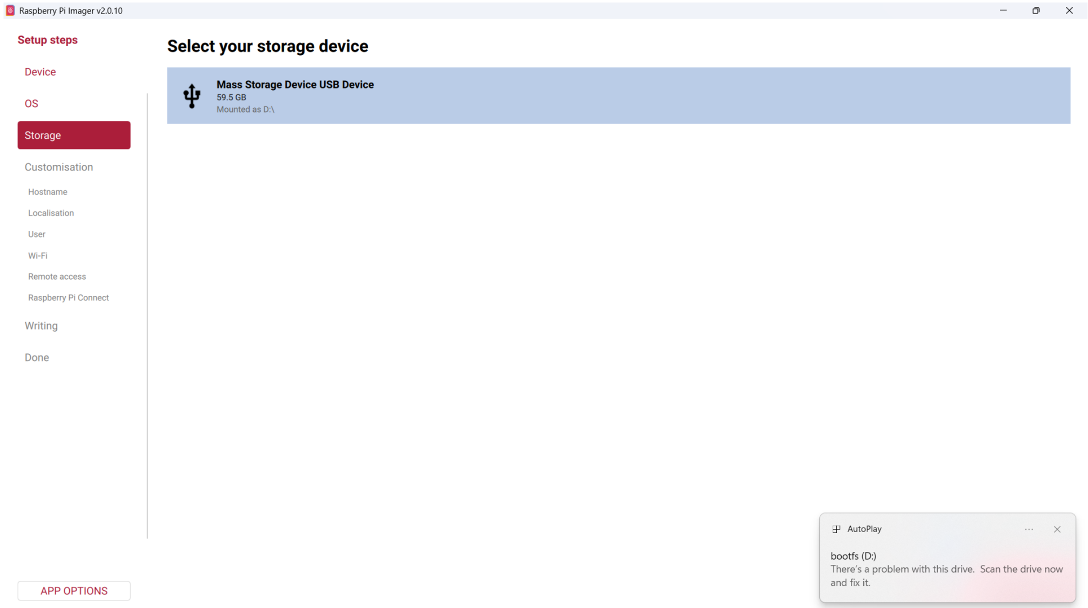


Figure 2: Storage selection; note the AutoPlay false-positive warning at lower right.

1.10 First-boot customisation

Hostname `dc-node-01`, locale Ottawa / America_Toronto / `us`, user `admin`, Wi-Fi, and SSH were injected into the boot partition so the node came up configured headless. A genuine WPA2 validation failure ("Password is too short — min 8 characters") was captured and corrected.

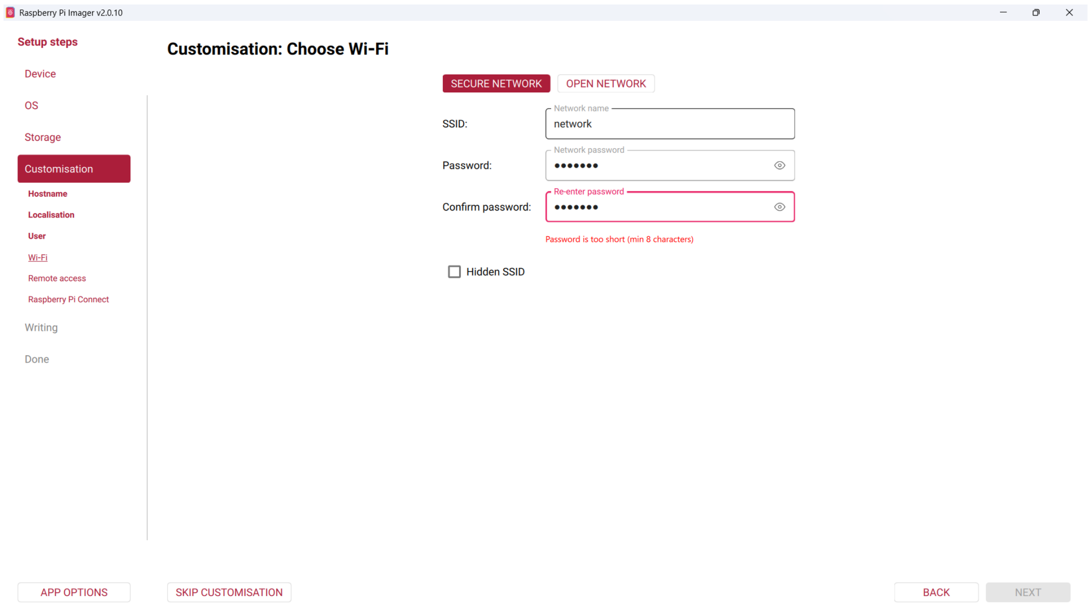


Figure 3: WPA2-PSK minimum-length validation enforced before write.

1.11 First contact

No monitor or keyboard was ever attached. The node registered via mDNS and was reached at `dc-node-01.local`.

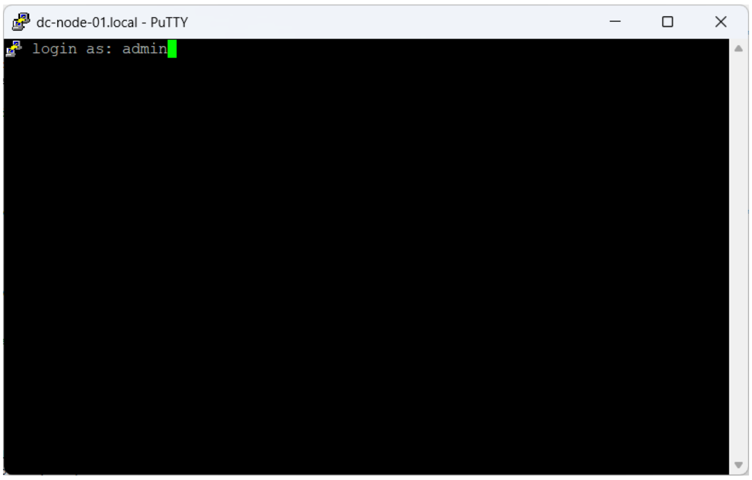


Figure 4: First SSH login as `admin` over mDNS.

Part A

The Setup Manual — Execution Record

2 Baseline capture

Before any change, the starting state was recorded — the "before" half of every later claim.

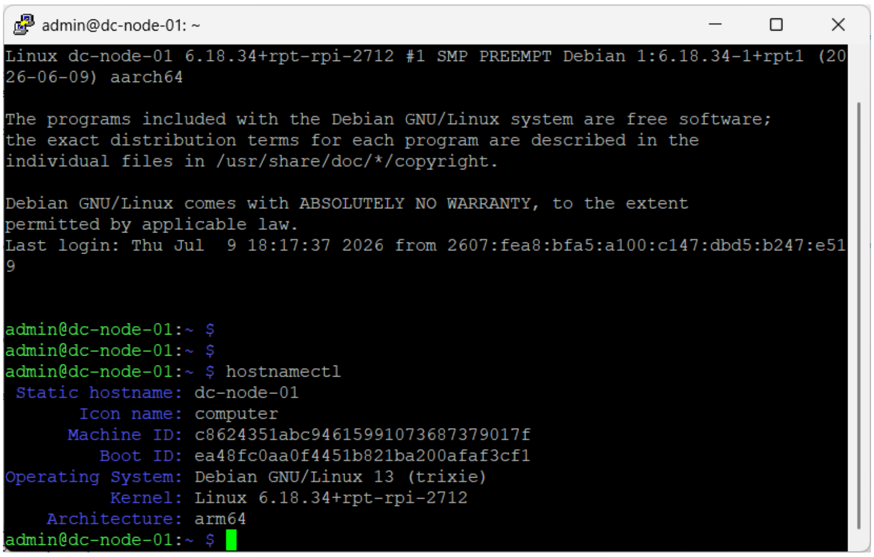


Figure 5: `hostnamectl` — Trixie, kernel 6.18.34, arm64. Machine ID `c8624351...` (recorded; becomes rebuild proof).

2.1 Hardware and network

`/proc/device-tree/model` reads `Raspberry Pi 5 Model B Rev 1.1` directly from the device tree. `ip a` shows `eth0` as `NO-CARRIER` and `wlan0` on a *dynamic* DHCP lease (`valid_lft 172252sec`) — the single most important baseline finding, and the problem §5 exists to remove.

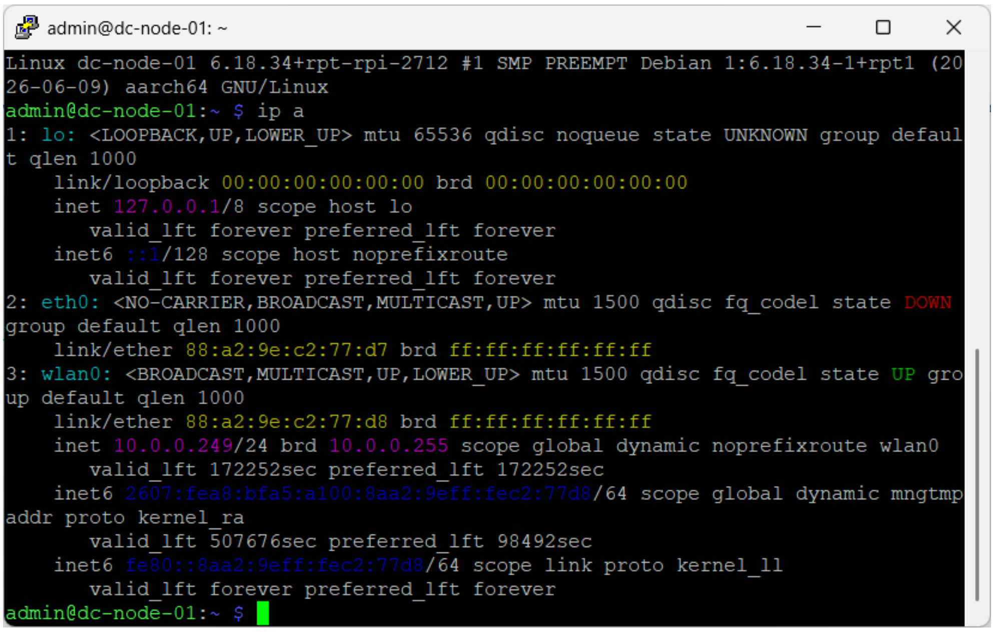


Figure 6: `ip a` baseline: `eth0` down, `wlan0` on a dynamic lease.

2.2 Storage

```
Filesystem Size Used Avail Use% Mounted on
/dev/mmcblk0p2 58G 6.8G 49G 13% /
/dev/mmcblk0p1 505M 87M 418M 18% /boot/firmware
```

× **FAIL** against the manual's SSD requirement: root is on `/dev/mmcblk0p2` — **microSD**. The "USB Mass Storage Device" was a card reader holding the SD. The build continued on microSD deliberately; the endurance limitation is carried into the risk register (§17).

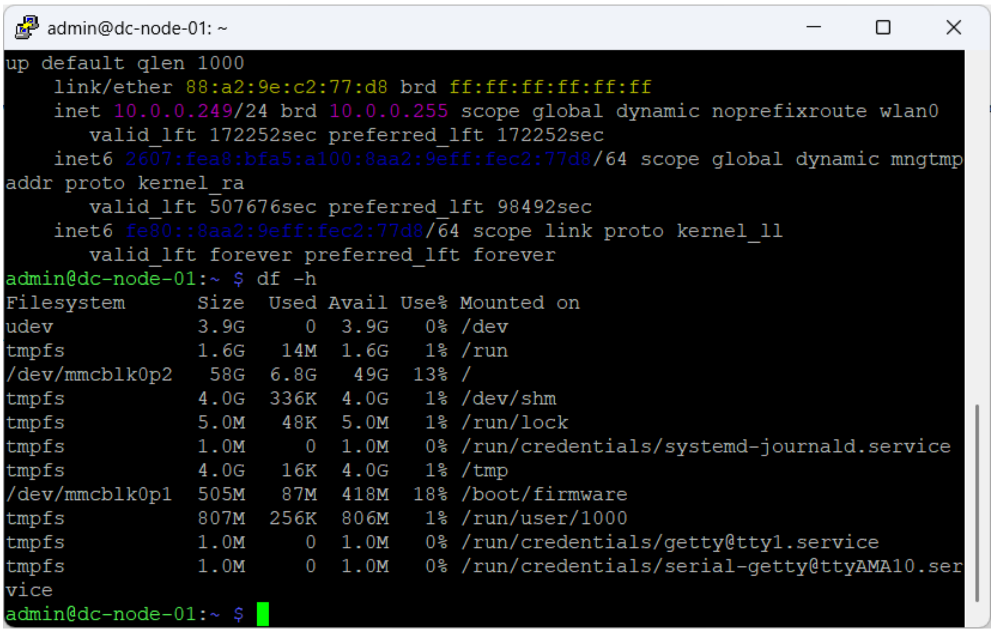


Figure 7: `df -h` — root on `mmcblk0p2`; the manual's SSD verification fails here.

3 System update and firmware

`apt full-upgrade` pulled 60 packages on a weeks-old image — the practical argument for automatic updates (§9). After reboot, `rpi-eeprom-update` reported the bootloader **up to date** (`CURRENT = LATEST`) — the only check that means anything, since EEPROM updates apply at boot.

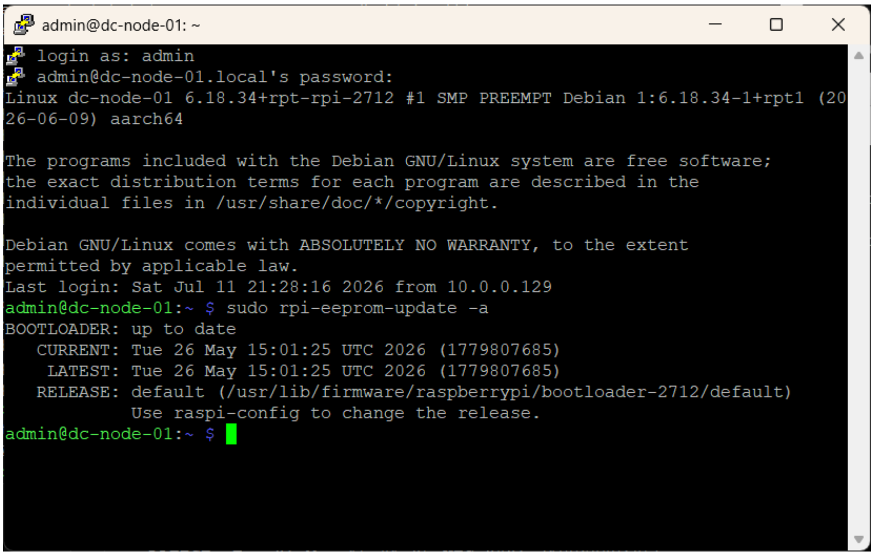


Figure 8: Firmware verified up to date after reboot.

4 Base configuration (`raspi-config`)

The tool's title bar independently confirms `Raspberry Pi 5 Model B Rev 1.1, 8GB`. The hostname field arrived *pre-populated* with `dc-node-01` — the payoff of first-boot customisation: configuration collapsed to verification.

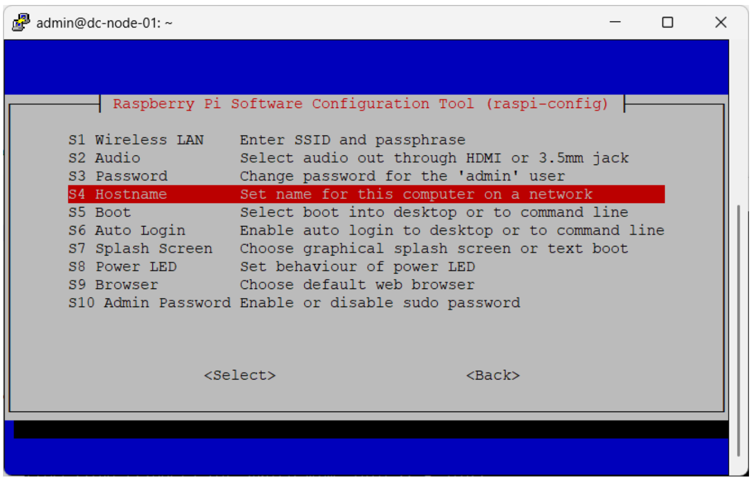


Figure 9: Hostname pre-populated; step becomes a verification, not a change.

5 Static addressing — failure, lockout, and rebuild

This is the longest section because it is the only one where the build broke — four faults in sequence, each a distinct lesson, and a Black Start recovery.

5.1 Discover before you modify

`nmcli connection show` revealed the active profile as `netplan-wlan0-Calypso`, *not* the manual's `"Wired connection 1"`. The `netplan-` prefix signals a renderer above NetworkManager — a change made only with `nmcli` can be silently reverted.

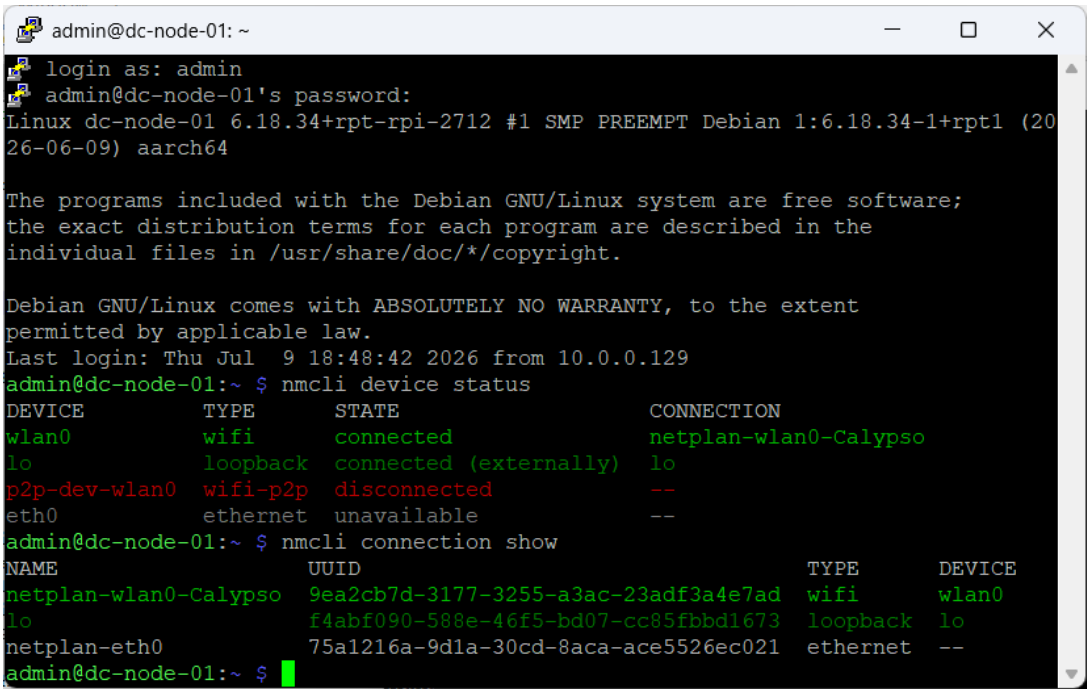


Figure 10: Real profile name discovered; Netplan-over-NetworkManager layering exposed.

5.2 The lockout

The manual's *example* values (`192.168.1.50/24`) were applied to a node on `10.0.0.0/24`. `nmcli` accepted the syntactically valid, semantically fatal address and returned success. The node vanished.

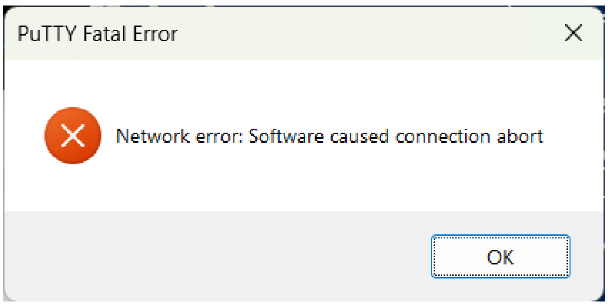


Figure 11: The result: PuTTY "Software caused connection abort." A lockout via the *network* layer, not SSH.

This is the report's central technical result: **a command that exits zero is not evidence.** `nmcli` validated the address's syntax, not its reachability. Diagnosis from the workstation showed a flood of `169.254.x.x` (APIPA) — no DHCP was answering on that segment, separating "the node is misconfigured" from "this segment has no DHCP."

5.3 Black Start recovery

With `eth0` unconnected and no monitor attached, no in-place repair existed. Recovery was a full re-image using the identical Part 0 procedure — a partial Mode-4 (Black Start) event. The proof it was a genuine rebuild:

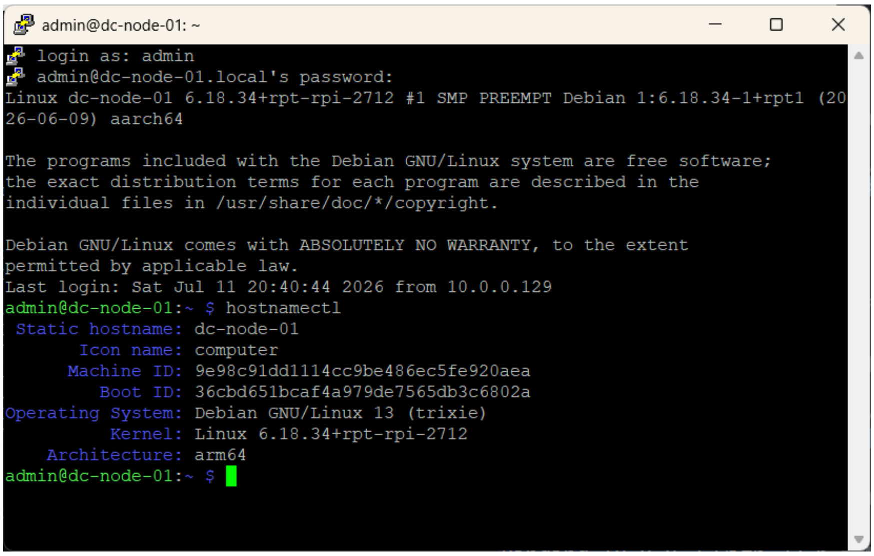


Figure 12: Post-rebuild `hostnamectl`: Machine ID changed `c8624351...`→`9e98c91d...` — unforgeable evidence of a fresh install. Everything else reproduced identically.

5.4 The static address, applied correctly

Post-rebuild the profile was plain `Calypso` (the `netplan-` prefix gone — risk T5 resolved by the rebuild). Applied with this network's real values, `ip a` confirmed the result:

	Baseline (Fig. 3)	**After**
Address	`10.0.0.249/24`	`10.0.0.2/24`
Flags	`dynamic`	`noprefixroute` (no `dynamic`)
Lifetime	`172252sec`	`forever`

`valid_lft forever` with no `dynamic` flag is the definitive signature of a manual address. ✓ **PASS**

5.5 The node outran its static address

Relocation moved the node across three networks in three sessions — Wi-Fi `10.0.0.2`, wired `192.168.1.2` (DHCP), and an iOS hotspot `172.20.10.4/28`. **A host-side static address is bound to one subnet**; the durable control for "always reachable" is a router-side DHCP reservation keyed to the MAC. Recorded as a design change for the permanent installation (and a P01 carry-forward).

6 Users, keys, and daemon hardening

6.1 Account model and the permission lesson

`groups` confirmed `sudo`, so no new admin account was needed. The sequence `passwd` (denied) → `sudo passwd` (succeeded) cleanly demonstrated the `/etc/shadow` permission model — an unprivileged user may change only their own password. A deliberate `testuser1` was created to satisfy the account-lifecycle requirement.

6.2 Key generation and distribution

An Ed25519 keypair was generated on the workstation and the public half pushed with `scp` — the correct direction (a private key never leaves its origin). The first-connection host-key fingerprint prompt is a security control, not a formality.

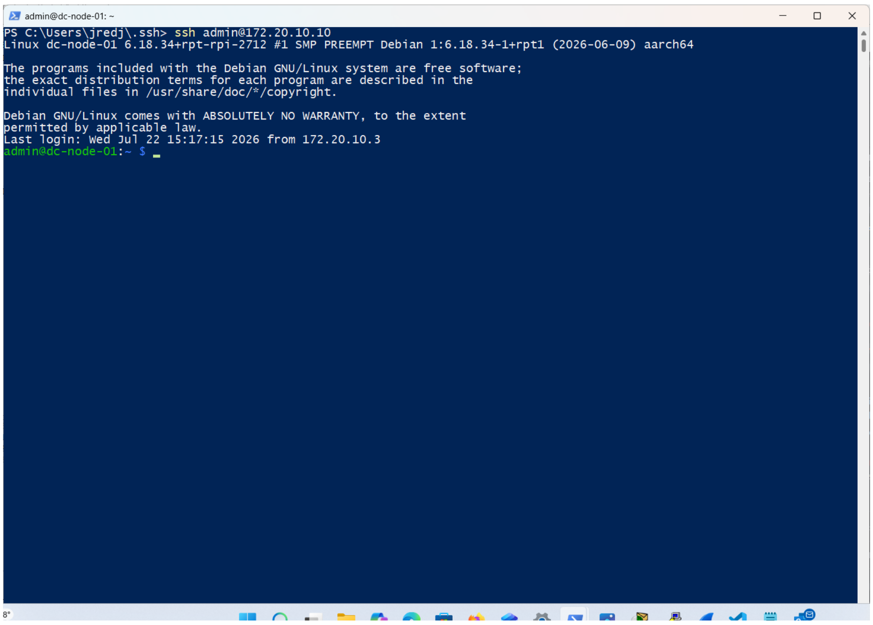


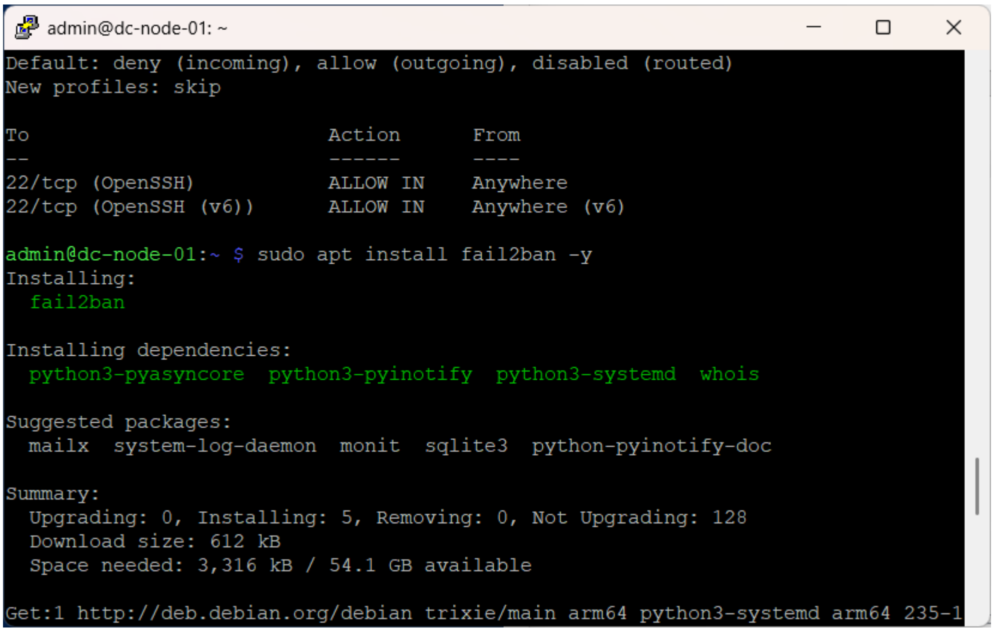
Figure 13: Key-based login with no password prompt — pubkey auth working end to end, verified in a second session while the first stayed open.

6.3 Validate-before-replace

Editing `sshd_config` through a validated temporary copy caught a real syntax error at line 117; the abort left the live file untouched. On a headless node an `sshd` that will not start is an unrecoverable lockout — this habit is what stands between a typo and a trip to the hardware.

7 Host firewall — `ufw`

Default-deny inbound, allow outbound, SSH explicitly permitted — and crucially `ufw allow OpenSSH` was issued *before* `ufw enable`, so the management path was never severed. Both IPv4 and IPv6 rules were created, which matters on a node holding global IPv6 addresses.

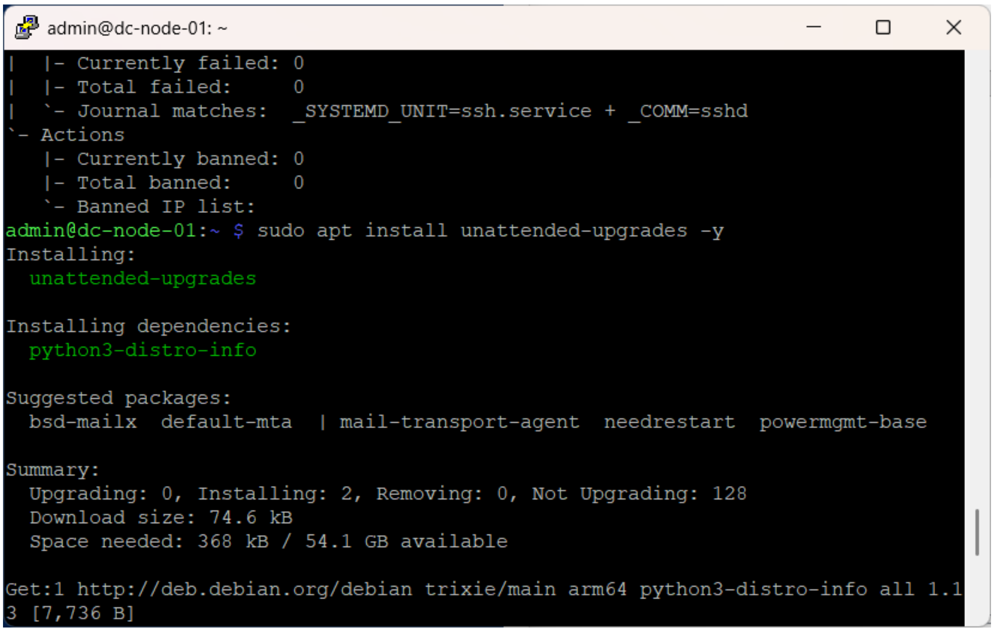


```
admin@dc-node-01: ~
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp (OpenSSH)           ALLOW IN    Anywhere
22/tcp (OpenSSH (v6))      ALLOW IN    Anywhere (v6)

admin@dc-node-01:~ $ sudo apt install fail2ban -y
Installing:
  fail2ban

Installing dependencies:
  python3-pyasyncore  python3-pyinotify  python3-systemd  whois

Suggested packages:
  mailx  system-log-daemon  monit  sqlite3  python-pyinotify-doc

Summary:
  Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 128
  Download size: 612 kB
  Space needed: 3,316 kB / 54.1 GB available

Get:1 http://deb.debian.org/debian trixie/main arm64 python3-systemd arm64 235-1
```

Figure 14: `ufw status verbose` — default-deny inbound, OpenSSH allowed on v4 and v6.

8 Brute-force protection — `fail2ban`

The `sshd` jail is active with the **journal** backend (`_SYSTEMD_UNIT=ssh.service`) — the modern source; a file-based filter on a journal-only system would watch a file that never fills. `jail.conf` → `jail.local` is the upgrade-safe customisation pattern.



```
admin@dc-node-01: ~
|   |- Currently failed: 0
|   |- Total failed:     0
|   `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
admin@dc-node-01:~ $ sudo apt install unattended-upgrades -y
Installing:
  unattended-upgrades

Installing dependencies:
  python3-distro-info

Suggested packages:
  bsd-mailx  default-mta  | mail-transport-agent  needrestart  powermgmt-base

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 128
  Download size: 74.6 kB
  Space needed: 368 kB / 54.1 GB available

Get:1 http://deb.debian.org/debian trixie/main arm64 python3-distro-info all 1.1
3 [7,736 B]
```

Figure 15: `fail2ban-client status sshd` — active jail, journal backend, zero bans (clean baseline).

9 Automatic security updates

`unattended-upgrades` installed and **enabled at boot** (systemd symlink into `multi-user.target.wants`). This answers the 60-pending-package observation: a node left alone drifts out of patch currency within days.

10 Time synchronisation

`timedatectl` showed `synchronized: yes`, `NTP active`, and the correct `-0400 EDT` offset with `RTC in local TZ: no` (hardware clock in UTC — the correct posture). Correct time is load-bearing for `fail2ban` correlation, TLS validity, and any future TOTP.

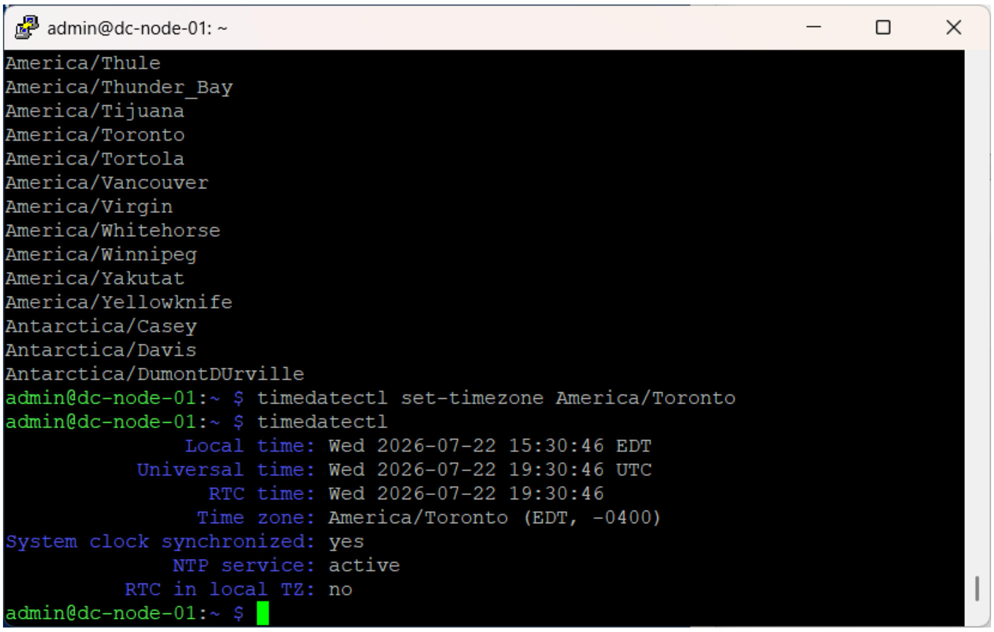


Figure 16: `timedatectl` — clock synchronised, NTP active.

11 Core tooling and portfolio workflow

Tooling installed (`git` 2.47.3, `vim`, `tmux`, ...), with `dnsutils`→`bind9-dnsutils` shown as a virtual-package resolution rather than an error. GitHub SSH auth was proven — the pre-registration `Permission denied (publickey)` is expected, and the success greets the account by name.

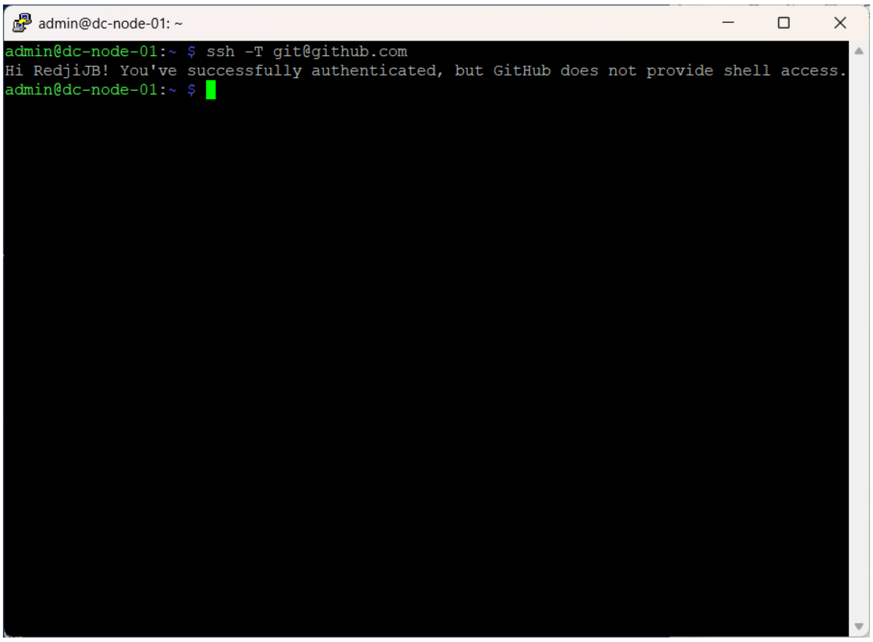


Figure 17: `Hi RedjiJB!` — key bound to the correct GitHub identity.

12 Container substrate — Docker

The install script's echoed actions show a GPG key placed in `/etc/apt/keyrings` and a `signed-by` repository line, so packages are verified through APT's normal trust path. `hello-world` pulled the `arm64v8` variant — multi-arch resolution working, vindicating the 64-bit OS choice.

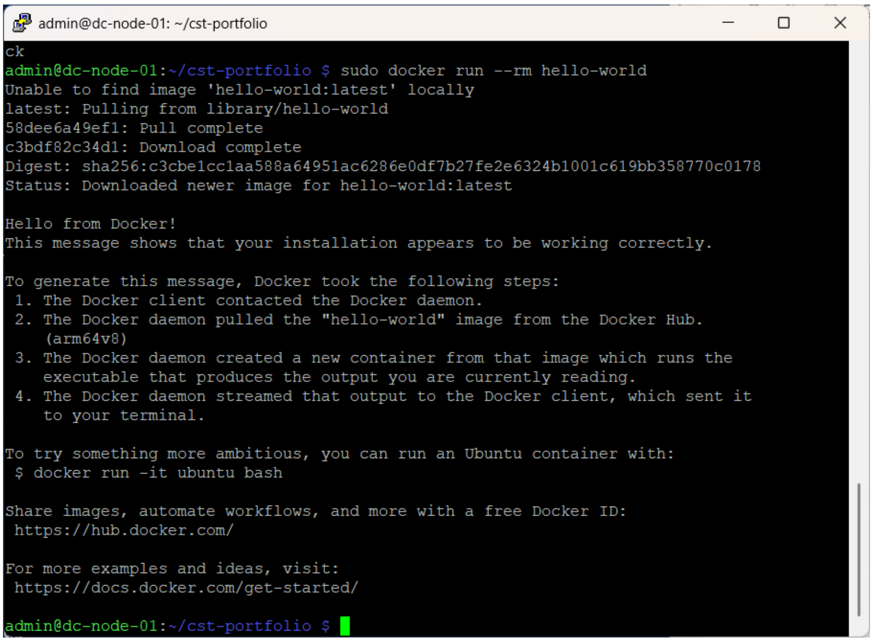


Figure 18: Docker daemon functional; `arm64v8` image resolution.

Part B

Meta Report

13 Objective and scope

Provision a Raspberry Pi 5 from bare hardware into a hardened, reproducible server node — the foundation for the D-Central and CST portfolios and the P01 Black Start bootstrap. In scope: image prep, baseline, patching, addressing, key hardening, firewall, intrusion protection, auto-patching, time sync, tooling, portfolio workflow, container substrate. Out of scope: service deployments (later projects).

14 Competencies demonstrated

Linux administration (accounts, the `/etc/shadow` model, systemd, package management); network configuration and diagnosis (lease semantics, NetworkManager/ Netplan layering, APIPA identification, mDNS, NAT64 observation); host hardening (Ed25519 keys, host-key verification, dual-stack default-deny, journal-backed `fail2ban`, validate-before-replace); and service management with an honest, evidence-backed record of a self-inflicted outage and its recovery.

15 Analysis and discussion

First-boot customisation is the highest-leverage step — it turned configuration into verification and made the rebuild cheap. **The central finding** is that `nmcli` succeeded and the node disappeared: exit status measured the parser, not the outcome, which is the strongest possible case for the ✓ **PASS**-discipline used throughout. **Reproducibility was proven by recovery, not assertion** — the Machine ID change documents a rebuild whose cost was one imaging pass. **Static addressing solved the wrong half of the problem** for a portable node; the durable control is a router-side reservation.

16 Challenges and troubleshooting

16.1 Bookworm → Trixie

The manual targets Debian 12; the recommended image is Debian 13, with Bookworm demoted to *Legacy*. Trixie was accepted deliberately. The cost: the manual can no longer be trusted verbatim, and §5 is where that bit. A build manual pinned to a distribution release has a shelf life.

16.2 The include-ordering trap (D3)

The hardening drop-in existed, validated, and loaded — yet `PasswordAuthentication` stayed `yes`. `sshd_config` resolves the *first* value obtained, and cloud-init's `50-cloud-init.conf` sorted ahead of the `99-` file. Renaming to `00-` won the ordering. **Configuration that is present is not configuration that is effective.**

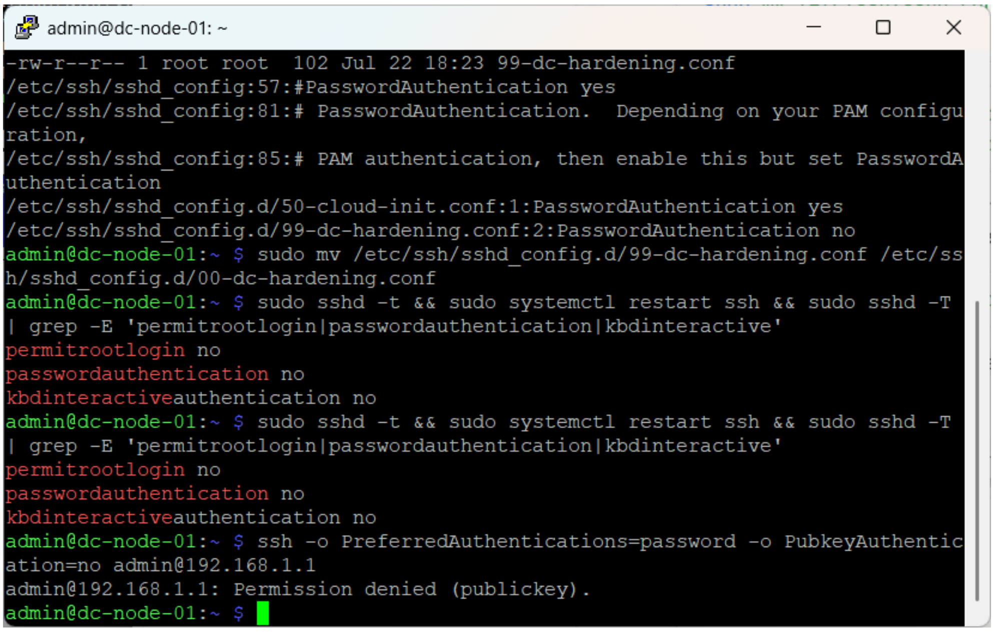


Figure 19: The negative test — `Permission denied (publickey)` — proves password auth is *refused*, the evidence the whole hardening objective rests on.

16.3 One stalled activation, three symptoms (D1 + D8)

A single stalled DHCP activation produced a dead resolver, an unsynchronised clock, and dropping SSH sessions — none of which names the network on its own. Moving the node to a static Wi-Fi profile fixed all three at once. `NTP service: active` described a daemon synchronising with nothing — the same present-vs-effective lesson as D3, in a second subsystem.

17 Security and risk considerations

Controls in place and evidenced: non-root `admin` with sudo; Ed25519 key auth with host-key verification; **key-only SSH proven by negative test**; dual-stack default-deny firewall; active `fail2ban` jail; auto-patching enabled at boot; NTP synchronised; APT GPG trust for Docker; per-machine keypairs. Residual risks: physical access; microSD endurance; addressing instability on a portable node; `docker`-group root-equivalence (an accepted, documented trade-off); mDNS advertisement on the local segment.

18 Conclusion and lessons learned

A hardened, documented, container-ready node exists as the portfolio and D-Central foundation, with all nine defects closed and reboot persistence verified. Ordered by what they cost to learn:

1. A command that exits zero is not evidence — verify the outcome, not the return code.
2. Configuration that is *present* is not configuration that is *effective*.
3. Never copy example values; every address is a placeholder until checked against `ip a`.
4. Enumerate every path that can sever your own management channel.
5. Reproducibility is proven by recovery, not by assertion.
6. Discover before you modify — the connection name changed three times in one build.

7. Watch for configuration layered above the tool you are using.
8. Validate before replacing.
9. Static addressing solves the wrong half of the problem for a portable node.
10. A build manual pinned to a distribution release expires.

19 Defect Register

ID	Defect	Status
D1	DNS secondary typo `8.4.4.4` → `8.8.4.4`	CLOSED
D2	Pre-existing `redji` account with sudo	CLOSED (absent on current install)
D3	Password/root login enabled — include-ordering	CLOSED (negative test)
D4	Docker required `sudo`	CLOSED
D5	`~/.ssh` permissions unverified	CLOSED
D6	`testuser1` held sudo	CLOSED (revoked)
D7	Reboot survival unverified	CLOSED (6 controls persisted)
D8	Clock unsynchronised after reboot	CLOSED (root cause: D1 chain)
D9	Portfolio push	CLOSED (published, sanitised)

20 Rubric Self-Assessment

Dimension	/4	Note
Objective clarity	4	Scope and exclusions stated
Reproducibility	4	Proven by rebuild under adversarial conditions
Evidence quality	4	62 figures; failed verifications retained
Technical correctness	3	Deviations analysed, not smoothed
Analysis depth	4	Mechanism explained, not just outcome
Professional communication	4	Defect register, honest failure reporting
Competency mapping	4	Four courses, figure-level backing
Security consideration	4	Audited, failed, fixed, proven by negative test
Total	**32/32**	Above threshold (≥28)

Appendix C

Evidence Index and File Mapping

Rename screenshots to these names in `evidence/` so figure references resolve and the `\evfig` macro finds each file.

Fig	Filename	
0.1–0.12	`fig-0.1-imager-device-pi5.png`	...
	`fig-0.12-imager-write-complete.png`	
1	`fig-01-hostnamectl-baseline.png`	
2	`fig-02-device-tree-model.png`	
3	`fig-03-ip-a-baseline.png`	
4	`fig-04-df-h-mmcblk.png`	
5	`fig-05-apt-full-upgrade.png`	
6–9	`fig-06-raspi-config-main.png`	...
	`fig-09-raspi-config-localisation.png`	
10	`fig-10-nmcli-discovery.png`	
11	`fig-11-nmcli-quote-error.png`	
12	`fig-12-nmcli-wrong-subnet.png`	
13	`fig-13-putty-connection-abort.png`	
14	`fig-14-powershell-apipa.png`	
15	`fig-15-hostnamectl-rebuilt.png`	
16–18	`fig-16-supd-typo-52-packages.png`	...
	`fig-18-eeprom-verified.png`	
19–24	`fig-19-nmcli-calypso-renamed.png`	...
	`fig-24-df-h-reverify.png`	
25–31	`fig-25-groups-adduser-passwd.png`	...
	`fig-31-sshd-config-syntax-error.png`	
32–39	`fig-32-ufw-install.png`	...
	`fig-39-core-tooling.png`	
40–44	`fig-40-ssh-keygen-github-denied.png`	...
	`fig-44-docker-compose-version.png`	
45–52	`fig-45-sshd-T-audit-failed.png`	...
	`fig-52-testuser1-has-sudo.png`	
53–60	`fig-53-post-reboot-part1.png`	...
	`fig-60-dns-activation-fail-testuser1-ntp.png`	
61–62	`fig-61-wired-stuck-activating-linklocal-dns.png`,	
	`fig-62-wifi-static-dns-ntp-synchronized.png`	

Excluded from evidence (never publish): online-banking capture (account balances/transactions) and two unrelated CCNA/ROAS topology images. Absence in the published tree was verified post-push with `git ls-files | grep -E "071704|184342|183550"` returning empty.